Problem Set 2

Industrial Organization

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Part 0: Empirical Target

Object of Interest

Our main empirical objective is to **quantify the welfare gains associated with the availability of gluten-free noodle options on college campuses**. Specifically, we aim to estimate the change in student welfare relative to a counterfactual market in which none of the chains offers gluten-free options. In addition to estimating the average welfare gain across students, we aim to characterize how these gains vary across student groups, particularly athletes and graduate students.

More precisely, our object of interest is the change in consumer surplus resulting from the introduction of gluten-free options, both on average and separately across groups defined by students' observable characteristics. This allows us to assess not only the overall benefits of expanding the available choice set, but also how these benefits are distributed across different types of students.

Demand Estimation and Consumer Preferences

To quantify these welfare gains, we will estimate a discrete-choice demand model based on a random utility framework. Student utility will depend on product prices and observable characteristics, such as spiciness, wheat content, the vegetable-to-meat ratio, and calories, as well as unobserved product and consumer characteristics.

We will begin with the logit specifications in Part II and then allow for heterogeneous preferences through random-coefficient models in Part III. These specifications will help us characterize how students trade off prices and product characteristics, as well as how preferences vary across students with different observable characteristics, particularly athletes and graduate students.

Price Variation and Product Substitution

A key challenge is that observed prices may be endogenous if stores adjust prices in response to unobserved product attractiveness. We will therefore investigate whether local input-cost shocks provide suitable instruments for product prices. Specifically, variation in local prices of wheat, vegetables, and meat, combined with differences in product composition, may generate exogenous variation in production costs and help identify price sensitivity.

We will also characterize substitution patterns across products. Observed market shares will inform the estimation of demand, while the hypothetical second-choice responses in the survey can provide

additional information on substitution between alternatives. These data may improve our understanding of how students would reallocate their choices if gluten-free products were unavailable.

Counterfactual and Welfare Measurement

Using the estimated demand model, we will simulate a counterfactual in which all gluten-free products are removed from the choice set, while the prices and characteristics of the remaining products are held constant. Holding the prices of the remaining products fixed is a simplifying assumption that isolates the direct welfare effect of expanding the available choice set and facilitates computation. It abstracts from potential equilibrium price adjustments following the introduction or removal of gluten-free products.

We will measure welfare changes using the model-implied change in consumer surplus, accounting for the estimated price sensitivity and, where applicable, heterogeneity in preferences. Finally, we will examine heterogeneity in welfare gains across student groups, particularly athletes and graduate students. Comparing results across the logit and random-coefficient specifications will allow us to assess how assumptions about preference heterogeneity affect the estimated benefits of gluten-free options.

Part 1: Empirical Evidence, Early Exhibits

Descriptive Exhibits

Figure 1 Panel A documents the evolution of noodle market shares between 2001 and 2020. Traditional noodles account for all of the noodle market before the introduction of gluten-free (GF) options in 2011. Following their introduction, GF noodles gain market share, with a further increase around 2016, when additional chains begin offering GF alternatives. Over the same period, the market share of traditional noodles declines. These patterns are consistent with a change in the composition of noodle demand as GF alternatives become available, although the figure alone cannot establish that their introduction caused the decline in traditional noodle shares. Panel B reproduces the same analysis by noodle chain. Since YourNoodles was the first to introduce GF options we observe its market share rising in 2011, but after all brands introduce it in 2016 it actually declines and others go up, suggesting BestNoodles and AllNoodles might have more attracting options at this point.

Related to that, Figure 2 examines whether GF choices vary across student groups over time. The share of GF choices increases for both athletes and non-athletes, as well as for graduate students and undergraduates since the first introduction of a GF option in 2011, particularly after 2015. Non-athletes seem to consistently consume more GF options, while for Graduate vs non Graduate-students this relationship varies depending on the year. The overall patterns suggest potential heterogeneity in adoption patterns. However, these comparisons are descriptive and may also reflect changes in the composition of the surveyed students and campuses over time.

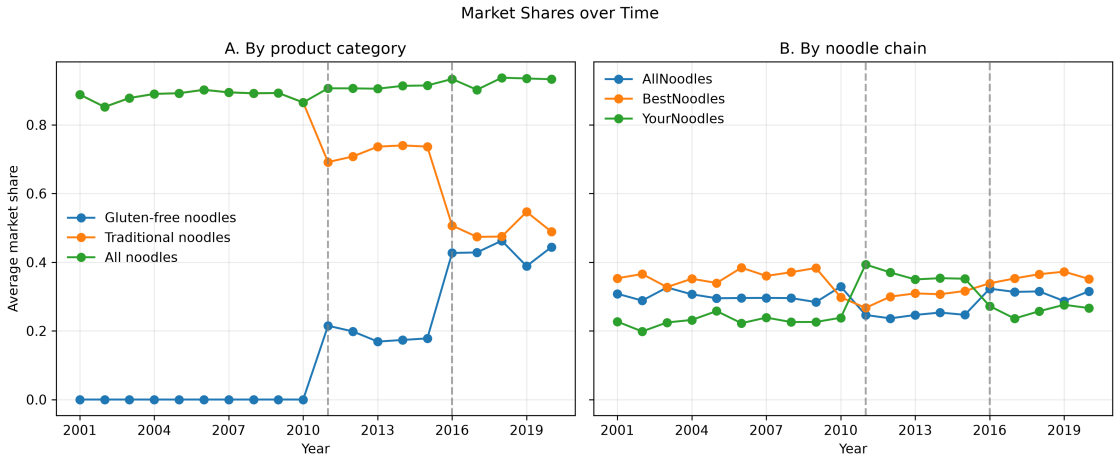


Figure 1: Market shares by product category and noodle chain

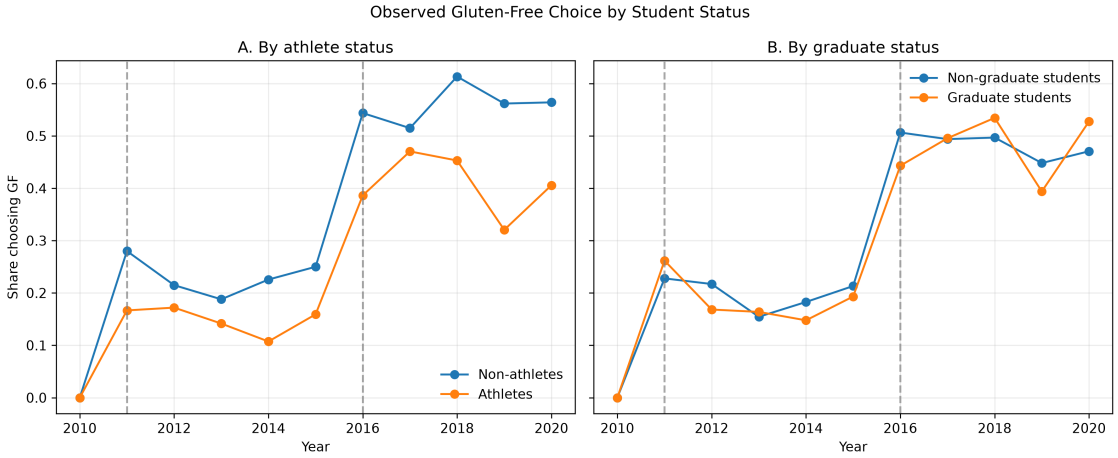


Figure 2: Observed gluten-free choice by athlete and graduate status

Pricing regressions

Now we procede to show results of pricing regressions. Our main exercises refer to Regressions 1 and 4 (baseline versus adding cost and market shocks) but we also add intermediate results adding both sets of shocks individually to better assess their effect in the estimation.

Regression 1 — Baseline

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + u_{jmt}.$$

Regression 2 — Market shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \gamma' C_{mt} + u_{jmt}.$$

Regression 3 — Input-cost shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \delta' Z_{jmt} + u_{jmt}.$$

Regression 4 — All cost shocks

$$Price_{jmt} = \beta_0 + X'_{jmt}\beta + \gamma' C_{mt} + \delta' Z_{jmt} + u_{jmt}.$$

Here, j identifies the product (brand and dish), m the campus, and t the year. The vectors enter the specifications, and appear in the regression table, in the following order:

- $X_{jmt} = (\text{SpicyIndex}, \text{WheatContent}, \text{Veg2meat}, \text{Calories})'$ contains the observed product characteristics.
- $C_{mt} = (\text{WageIDX}, \text{ReEsIDX})'$ contains market-level cost shifters: the local wage and real-estate indices.
- $Z_{jmt} = (\text{wheat_shock}, \text{vegetable_shock}, \text{meat_shock})'$ contains product-market-specific input-cost shocks, constructed from local input prices and the product's exposure to wheat, vegetables, and meat.

The vectors β, γ, and δ are the corresponding coefficient vectors, and u_{jmt} is the unobserved component of price. Standard errors are heteroskedasticity robust (HC1), so all estimates are interpreted as conditional associations rather than causal effects.

Table 1 reports regression results. Column (1) shows the OLS relationship between menu prices and product characteristics. Since prices may be correlated with unobserved product characteristics, we interpret the coefficients as conditional associations rather than causal effects. Wheat content is positively and statistically significantly associated with price, whereas the coefficients on spiciness, vegetable-to-meat ratio, and calories are not statistically significant. Advancing to columns (2) and (3) regression results suggest that both market-level cost shifters and product-market-specific input-cost shocks contribute to explaining price variation. Adding wage and real-estate indices increases the R-squared from 0.055 to 0.163, while adding input-cost shocks alone raises it to 0.308. Both sets of shocks are jointly statistically significant at the 1% level. When included together, the R-squared reaches 0.404. These results suggest that product-market-specific input-cost shocks account for a larger share of the variation explained by the pricing regressions than market-level cost shifters alone. The increase in explanatory power when both sets are included also suggests that they provide complementary information about prices.

However, these results establish explanatory relevance in the pricing regressions, not instrument strength in the demand estimation or instrument exogeneity. The latter requires further assessment in the instrumental-variables estimation and an economic justification for the exclusion restrictions.

Table 1: Pricing Regressions with Market and Input-Cost Shocks

Variable	(1) Baseline	(2) Market shocks	(3) Input-cost shocks	(4) All cost shocks
const	0.6768***	0.2821	0.7605***	0.4085***
	(0.1894)	(0.1894)	(0.1547)	(0.1504)
SpicyIndex	-0.0622	-0.0463	-0.0264	-0.0111
	(0.0630)	(0.0608)	(0.0519)	(0.0496)
WheatContent	0.7055***	0.7309***	-0.1676**	-0.1235*
	(0.0699)	(0.0651)	(0.0696)	(0.0638)
Veg2meat	0.0755	0.0492	-0.3255***	-0.3531***
	(0.1209)	(0.1155)	(0.1218)	(0.1151)
Calories	-0.0225	0.0038	-0.0838	-0.0546
	(0.2088)	(0.2017)	(0.1669)	(0.1604)
WageIDX	—	0.1753***	—	0.1688***
		(0.0200)		(0.0199)
ReEsIDX	—	0.0547***	—	0.0338**
		(0.0196)		(0.0151)
wheat_shock	—	—	0.6815***	0.6658***
			(0.0319)	(0.0293)
vegetable_shock	—	—	0.2351***	0.2341***
			(0.0481)	(0.0472)
meat_shock	—	—	-0.0031	-0.0035**
			(0.0019)	(0.0017)
Observations	2150	2150	2150	2150
R-squared	0.0550	0.1625	0.3082	0.4039
F-statistic (included shocks)	—	40.57	161.39	132.19
F-test p-value	—	<0.001	<0.001	<0.001

Notes: Dependent variable: price. Heteroskedasticity-robust standard errors are in parentheses. The F-test tests whether the shocks included in each specification are jointly equal to zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Reduced-Form Estimations

For our reduced form exhibits we begin wit Figure 3 which reports the event-study estimates around the introduction of gluten-free options. The following figures then describe market shares overall and heterogeneity in observed gluten-free choices.

The event-study estimates show no statistically significant joint evidence of pre-adoption differences relative to the reference period ($F = 1.255$, $p = 0.324$). In contrast, the post-adoption coefficients are jointly statistically significant ($F = 10.860$, $p < 0.001$), indicating changes in chain-level market shares following gluten-free introductions. These results are suggestive of a relationship between product introduction and market outcomes, but they do not establish a causal effect given the small number of adopting chains and potential spillovers across competitors.

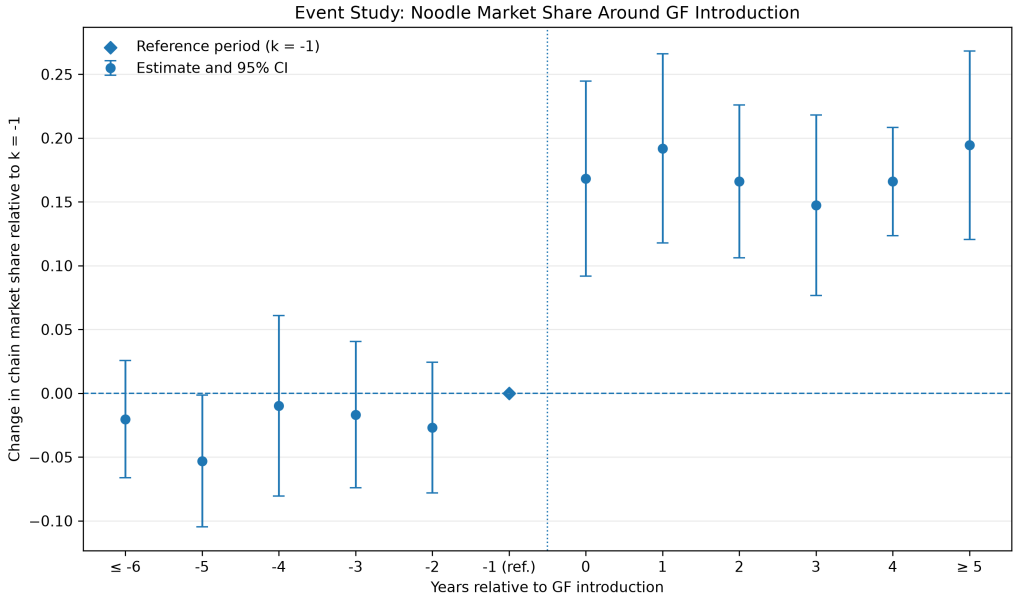


Figure 3: Event study: noodle market share around gluten-free introduction

On another exhibit, Table 2 documents substitution patterns across chains and product categories. Among respondents whose observed choice was gluten-free, the share reporting a traditional noodle dish as their hypothetical second choice ranges from 54.7% for AllNoodles to 80.8% for YourNoodles. Conversely, among respondents whose observed choice was traditional, 24.5%–31.8% report a gluten-free dish as their second choice, depending on the chain. These patterns suggest that gluten-free and traditional noodles can be substitutes, but the extent of substitution varies across chains. They also indicate that product category alone does not fully characterize consumers' second-choice patterns. Since these are stated hypothetical choices, they should not be interpreted as actual switching behavior or as direct estimates of substitution following the removal of all gluten-free products.

Table 2: Second-Choice Summary by Chain

Observed choice	Second choice: GF	Second choice: Traditional	N
AllNoodles – GF	45.3%	54.7%	752
AllNoodles – Traditional	31.8%	68.2%	2,382
BestNoodles – GF	39.4%	60.6%	836
BestNoodles – Traditional	27.3%	72.7%	2,750
YourNoodles – GF	19.2%	80.8%	1,527
YourNoodles – Traditional	24.5%	75.5%	1,784

Part 2

5.

Dividing everything by σ,

$$\frac{1}{\sigma}\, u_{ijmt} = -\frac{1}{\sigma}\, p_{jmt} + \frac{\mu_j}{\sigma} + \frac{\beta'}{\sigma} X_{jmt} + \frac{\xi_{jmt}}{\sigma} + \varepsilon_{ijmt}.$$

If we define $\alpha = 1/\sigma$ and rescale the coefficients and utility, we get

$$\tilde{u}_{ijmt} = -\alpha\, p_{jmt} + \tilde{\mu}_j + \tilde{\beta}' X_{jmt} + \tilde{\xi}_{jmt} + \varepsilon_{ijmt}.$$

From now on, this is the form we use, dropping the tildes to have everything rescaled.

6.

First, let $\delta_{jmt} = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}$. Choice probabilities are

$$s_{jmt} = P\Big\{\delta_{jmt} + \varepsilon_{ijmt} \geq \delta_{\ell mt} + \varepsilon_{i\ell mt}\ \forall \ell\Big\} = \frac{e^{\delta_{jmt}}}{\sum_\ell e^{\delta_{\ell mt}}},$$

by the properties of the Gumbel distribution. In the data, we see that the shares add up to less than one. We consider that those who are not consuming noodles are picking the outside option of good 0, which has utility ε_{i0mt}, and mean utility $\delta_{0mt} = 0$. By the same logit formula, the outside option has market share $1/(\sum_\ell e^{\delta_{\ell mt}})$. Dividing one by the other,

$$\frac{s_{jmt}}{s_{0mt}} = e^{\delta_{jmt}} \implies \log(s_{jmt}) - \log(s_{0mt}) = \delta_{jmt}.$$

Using the definition of δ, we get the linear regression

$$\log(s_{jmt}) - \log(s_{0mt}) = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}.$$

Under the assumption that $\mathbb{E}\Big[\xi_{jmt} \mid p_{jmt}, X_{jmt}\Big]$, the parameters can be unbiasedly recovered using OLS. As each good has a different μ_j, we include good fixed effects. Estimates are in the first column of the table below.

Table 3: Demand Estimates

	OLS	IV	
Model:	(1)	(2)	(3)
Variables			
Price	-0.638***	-0.826***	-0.829***
	(0.006)	(0.013)	(0.011)
Spicy	-0.063	-0.047	-0.047
	(0.079)	(0.093)	(0.093)
Wheat	0.144***	0.268***	0.270***
	(0.051)	(0.061)	(0.061)
Veg2Meat	-0.029	-0.020	-0.019
	(0.036)	(0.042)	(0.043)
Calories	0.134*	0.155*	0.156*
	(0.072)	(0.084)	(0.084)
Cost-shifter IVs	No	Yes	Yes
Interaction IVs	No	No	Yes
Fixed-effects			
dish	Yes	Yes	Yes
Fit statistics			
Observations	2,800	2,800	2,800
R^2	0.82842	0.45422	0.56939
Within R^2	0.81054	0.39738	0.52454

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

7.

Now we assume that the characteristics are exogenous, but that prices can be exogenous. We had to do some data cleaning to get all the university names to match in both datasets. We use the variables in Z as instruments for price. To obtain more variation, we also add interactions between Z and how important it is for a good (for example, Wheat Content $\times$ Wheat Costs). This is so that we also have variation across universities, and not only over time. Results are in the same table as before.

As we would expect prices to be positively correlated to positive demand shocks, the price coefficient in OLS should have an upward bias, being closer to zero. This is exactly what we see: the price coefficient becomes more negative with instruments.

8.

Consumer surplus for a given scenario, at a given university and time, takes the form

$$\text{CS} = \frac{1}{\alpha} \log \left(\sum_{\ell} e^{\delta_{\ell mt}} \right).$$

These can be easily computed by plugging in the $\hat{\delta}$'s, which are the predicted values from the regressions. We simulate a counterfactual where gluten-free options were never introduced. We assume that, in that world, prices and everything else would have remained constant. We create a function that takes in the data and a vector or parameters, and outputs the welfare of each university and time. For the counterfactual, we simply drop gluten-free alternatives.

We see, in the figure below, that, while the counterfactual consumer surplus remains flat, the actual estimated one has two increases. The first is in 2011, when gluten-free options started to be introduced, and the second in 2016, when more options became available. The same pattern holds for OLS and for IV, but the scale is different.

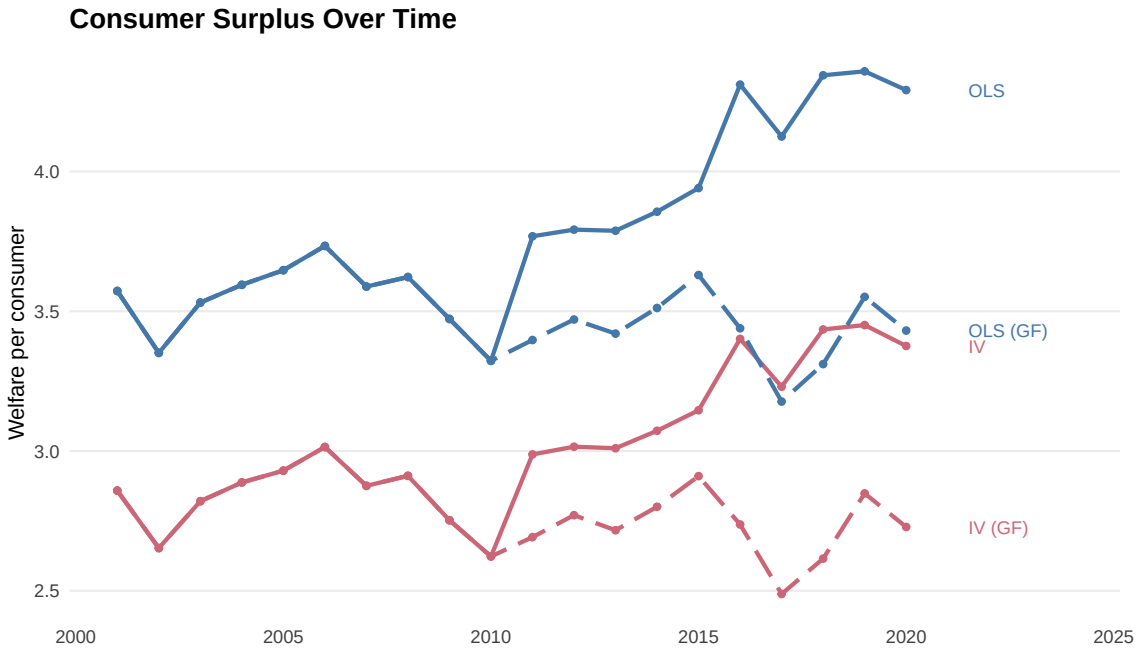


Figure 4: Consumer welfare over time

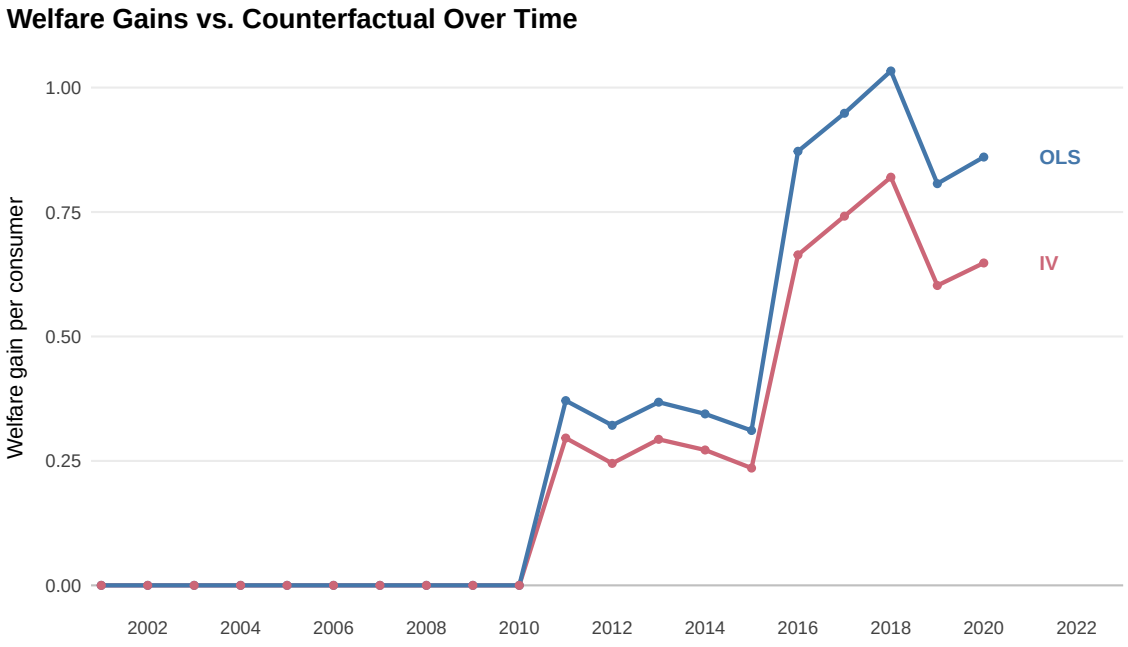


Figure 5: Welfare gains over time

Regressing the welfare gains on the characteristics of each university per year, we cannot find any heterogeneity depending on the type of student population.

Table 4: Welfare Gains and Student Composition

	2011–2015	2016 onward
Panel A: OLS		
Constant	0.341***	0.731***
	(0.065)	(0.122)
Share of Athletes	-0.008	-0.013
	(0.073)	(0.123)
Share of Graduates	0.013	0.364*
	(0.113)	(0.198)
Panel B: IV		
Constant	0.263***	0.522***
	(0.063)	(0.116)
Share of Athletes	-0.015	0.003
	(0.070)	(0.117)
Share of Graduates	0.026	0.348*
	(0.109)	(0.189)

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9.

The elasticities implied by logit are

$$\eta_{jk} = p_k \frac{\partial \log s_j}{\partial p_k} = p_k \frac{\partial}{\partial p_k} \left[\delta_j - \log \sum_\ell e^{\delta_\ell} \right].$$

So, elasticities are

$$\eta_{jk} = p_k \Big[-\alpha \mathbb{I}\{j = k\} + s_k \Big].$$

They depend only on data and on our estimate $\hat{\alpha}$. We compute the elasticity for each university and time, then average them out, using both the OLS and IV estimates. Below are our estimates of average elasticities, for the three time periods.

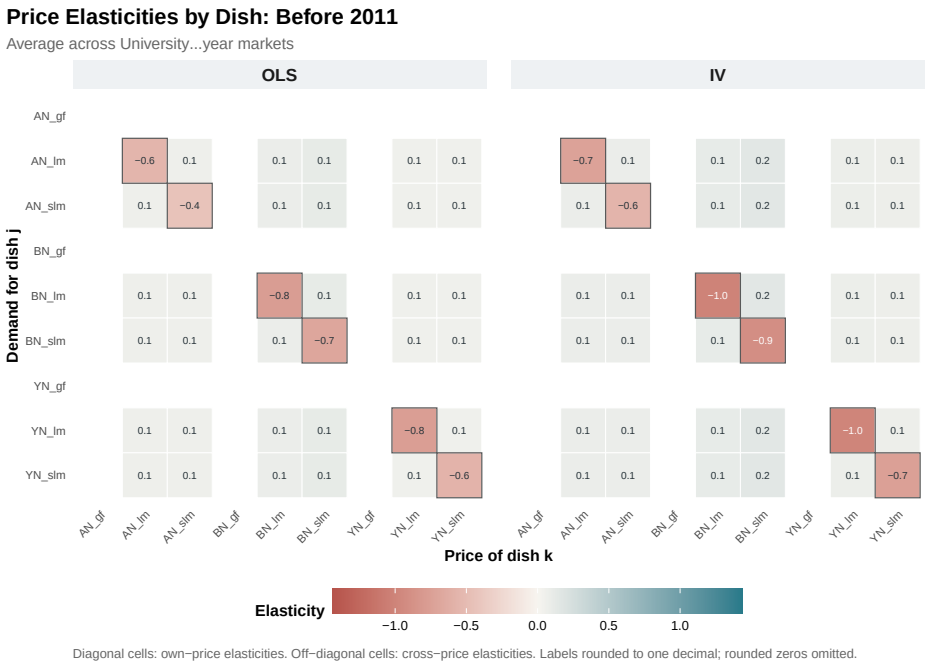


Figure 6: Price elasticities, before 2011

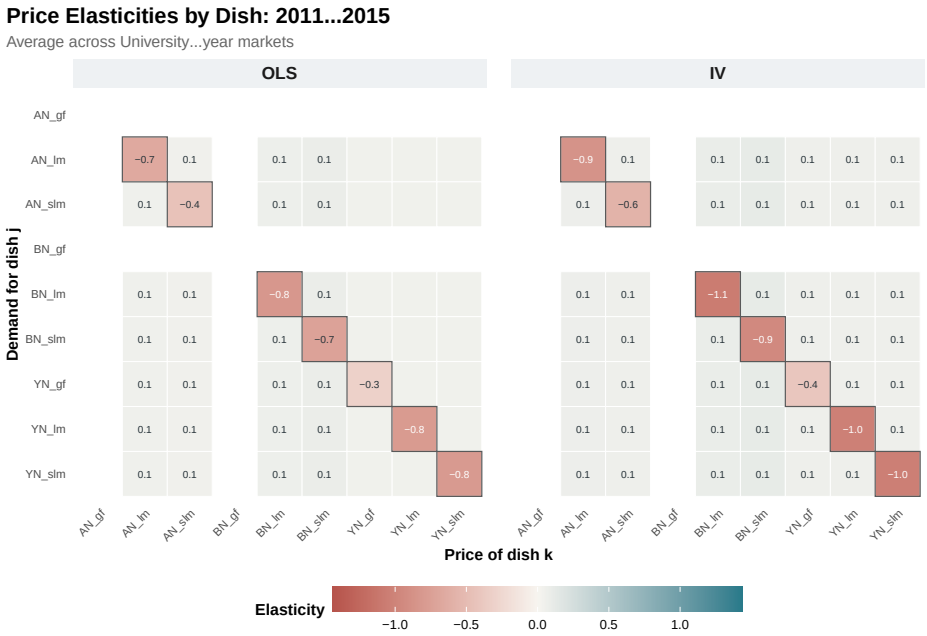


Figure 7: Price elasticities, 2011-2015

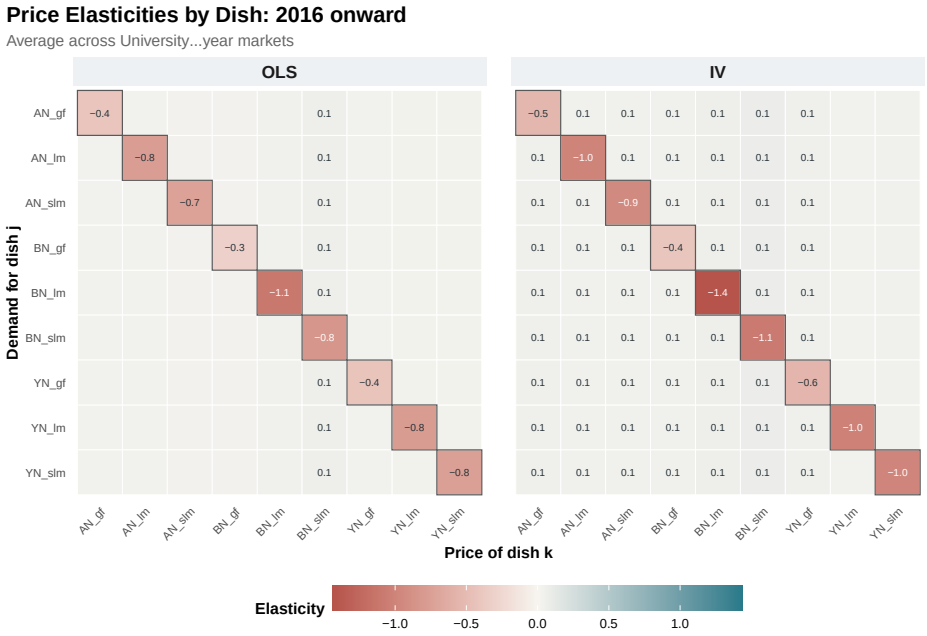


Figure 8: Price elasticities, 2016 onward

Because the market is relatively evenly split and the logit elasticities depend only on market shares and not characteristics, most of the cross-price elasticities sit at around 0.1. This is not intuitive, as the gluten-free options should be closer substitutes among each other, for example. In terms

of policy, this implies that the one company producting gluten-free noodles between 2011 and 2015 did not actually hold much market power, as consumers would be willing to easily substitute between gluten-free and gluten-containing dishes. Given this demand, a counterfactual where a single company became a monopolist of gluten-free options likely would not have a large spike in prices, as that would push consumers towards the other alternatives.

10.

Allowing for there to be two nests, $g = 0, 1$, we use the utility specification

$$u_{ijmt} = \delta_{jmt} + \zeta_{imt} + (1 - \rho)\varepsilon_{ijmt}, \quad \text{for } j \neq 0$$

where $\delta_{jmt} = -\alpha\, p_{jmt} + \mu_j + \beta' X_{jmt} + \xi_{jmt}$, and the outside option, in a different nest, has utility $u_{i0mt} = \varepsilon_{i0mt}$. The resulting expression for market shares is

$$\log(s_j) - \log(s_0) = \delta_{jmt} + \rho\, \log(s_{j|1}), \quad \text{where } s_{j|1} = \frac{s_j}{1 - s_0}.$$

This means we get a similar regression specification, with one extra term. Besides the cost-shifters we had before, we also add a third specification where we add the number of other elements in the next as an instrument.

Table 5: Nested Logit Demand Estimates

	OLS		IV	
Model:	(1)	(2)	(3)	(4)
Variables				
Price	-0.477***	-0.861***	-0.813***	-0.838***
	(0.008)	(0.032)	(0.021)	(0.016)
ρ	0.355***	-0.120	0.039	-0.022
	(0.014)	(0.100)	(0.044)	(0.027)
Spicy index	-0.067	-0.047	-0.047	-0.046
	(0.071)	(0.095)	(0.092)	(0.094)
Wheat content	0.116**	0.265***	0.268***	0.271***
	(0.046)	(0.062)	(0.060)	(0.061)
Vegetable-to-meat	-0.038	-0.018	-0.020	-0.019
	(0.033)	(0.043)	(0.042)	(0.043)
Calories	0.101	0.164*	0.152*	0.158*
	(0.065)	(0.087)	(0.084)	(0.085)
Cost-shifter IVs	No	Yes	Yes	Yes
Interaction IVs	No	No	Yes	Yes
Nest-size IV	No	No	No	Yes
Fixed-effects				
dish	Yes	Yes	Yes	Yes
Fit statistics				
Observations	2,800	2,800	2,800	2,800
R^2	0.86088	0.45435	0.56945	0.56920

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Addings instruments, ρ seems to be very close to 0, which means that the model collapses to a standard logit, and the nest doesn't really help. If we run the welfare exercises again for a ρ very close to zero, the results will be basically the same as in the standard logit case. Instead, we use the naive OLS estimates, since it is the only one with a ρ that is significant and positive. Here, the welfare also depends on ρ:

$$\text{CS} = \frac{1}{\alpha} \log \left(1 + \left(\sum_{\ell \neq 0} e^{\delta_{\ell m t}/(1-\rho)} \right)^{1-\rho} \right).$$

And the elasticities are

$$\eta_{jk} = \alpha p_k \left[s_k + \frac{\rho}{1-\rho} s_{k|1} - \frac{1}{1-\rho} \mathbb{I}\{j = k\} \right].$$

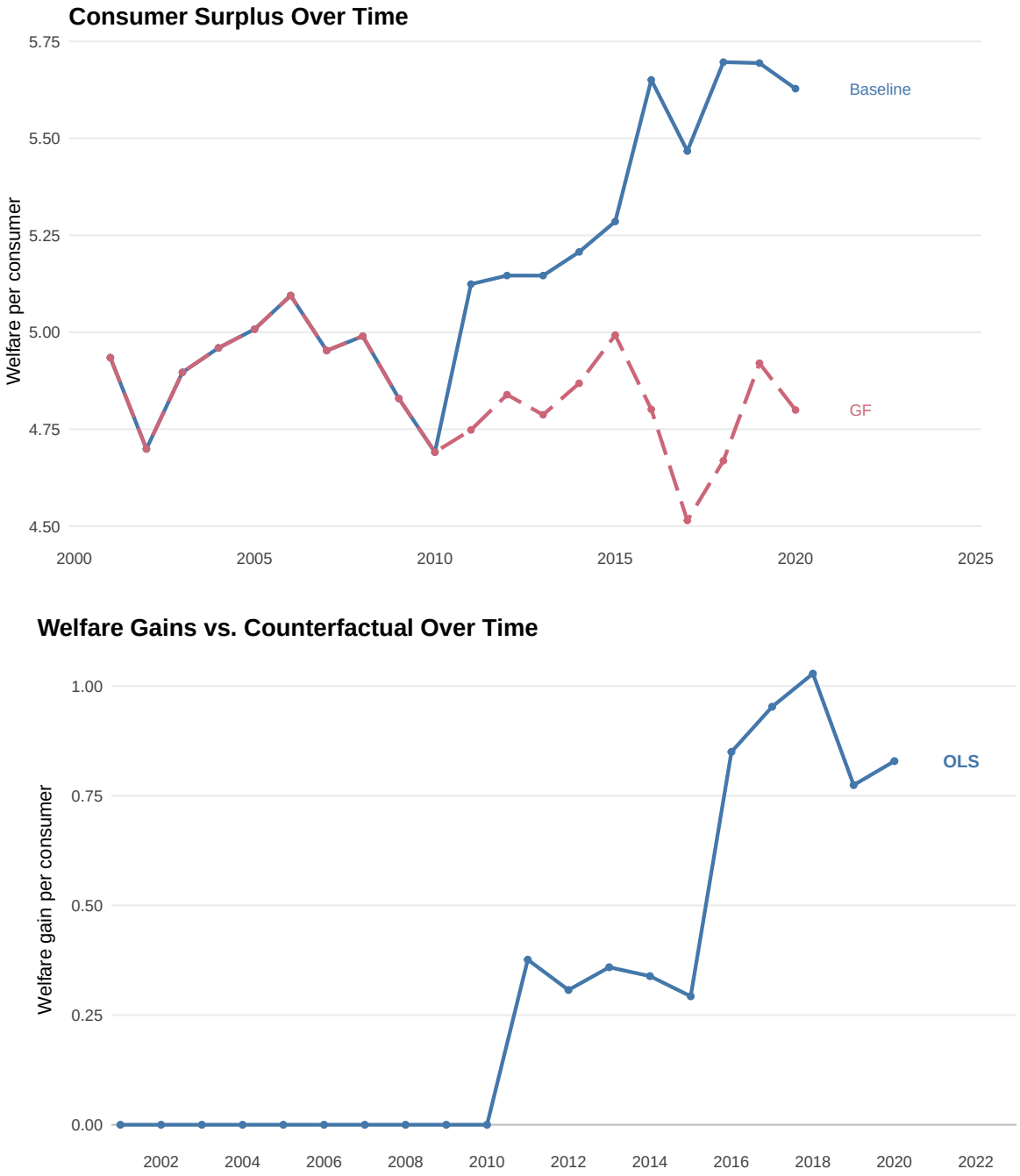


Figure 9: Welfare gains over time

Table 6: Welfare Gains and Student Composition

	2011–2015	2016 onward
Constant	0.331***	0.670***
	(0.078)	(0.138)
Share of Athletes	−0.016	0.020
	(0.088)	(0.139)
Share of Graduates	0.026	0.420*
	(0.136)	(0.225)

Dependent variable: nested-logit welfare gain. OLS demand estimates are used to construct welfare. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

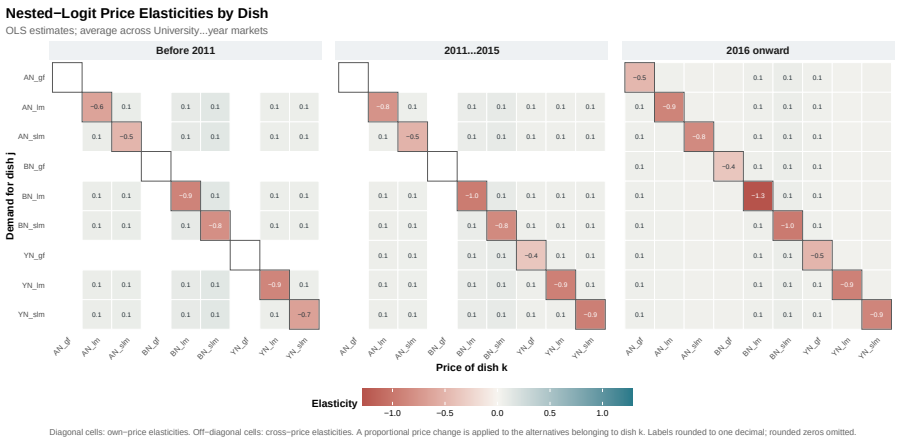


Figure 10: Elasticities

Overall, results are very similar to using the OLS estimator for standard logit.

11.

The Veg2Meat variable fluctuates a lot within dishes, and none of the dishes always have a ratio of 1. Therefore, none of the dishes are always vegetarian. Because of this, we use three nests: one for the gluten-free goods, one for the rest of the noodles, and one for the outside option. For simplicity, we focus on the specification with the most instruments. Here, the regression specification is

$$\log(s_j) - \log(s_0) = \delta_{jmt} + \rho \log(s_{j|g(j)}), \quad \text{where } s_{j|g} = \frac{s_j}{\sum_{\ell \in g(j)} s_\ell},$$

and the outside option just has utility ε_{i0mt}.

Table 7: Nested Logit Demand: Three Nests

Model:	(1)
Variables	
Price	-0.830***
	(0.015)
ρ	-0.003
	(0.041)
Spicy index	-0.047
	(0.093)
Wheat content	0.270***
	(0.061)
Vegetable-to-meat	-0.019
	(0.043)
Calories	0.156*
	(0.085)
Cost-shifter instruments	Yes
Interaction instruments	Yes
Nest-size instrument	Yes
Fixed-effects	
dish	Yes
Fit statistics	
Observations	2,800
R^2	0.56932

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The ρ we find is, once again, very close to zero, which means the ensts add very little information. The welfare exercises will yield pretty much the same results as we had before. The price coefficient is also very close to the one we had for simple logit.

Part 3